LESSON PLAN

EMI/EMC (14)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	25
Total No. of Lesson Plans	46	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Explanation of leveling and compensation system
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Explanation of leveling and compensation system.
5. Specific Objectives	14.2.1.1 Trainee will be able to explain the functions of AFCS.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of AFCS schematic diagrams; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the AFCS?
(b) Why is aircraft leveling necessary during flight?
(c) How does the compensation system correct attitude errors?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Basic principles of aircraft leveling • Function of artificial horizon sensors • Role of compensation system in AFCS • Correction of bank and pitch errors • Integration of leveling data with autopilot	L & Ds	PPT, IB	15 min
(b)	• Mechanism of pendulous vanes operation • Electrical compensation circuit logic • Impact of turbulence on leveling accuracy • System response to rapid attitude changes • Maintenance checks for leveling sensors	L & Ds	PPT, PDF	20 min
(c)	• Troubleshooting common leveling faults • Calibration procedures for AFCS units • Interpretation of AFCS warning indicators • Effect of magnetic interference on sensors • Documentation of system performance logs	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the compensation system corrects false horizon indications.
(b) Describe the relationship between pendulous vanes and AFCS stability.

12. Home Assignment and Follow Up	Prepare a brief note on the transition from LP 25 leveling systems to LP 26 side slip sensing mechanisms, highlighting sensor location differences.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Side slip sensing and turn co-ordination, location of sensing elements in the A/C and explanation of inertial attitude references.	Refer to the master syllabus reference list	LP 26

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min